



Research paper

Nonlinear identification of surface ship manoeuvring motion model and its control application

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ABSTRACT

This paper develops and investigates a square root unscented Kalman filter (SRUKF) for motion state prediction and parameter estimation of surface ship manoeuvring motion model. Based on the identified model and adaptive sliding mode control, trajectory tracking control of the surface ship is conducted with consideration of ship maneuverability. To balance accuracy, complexity, and practical control applications, this paper establishes a 3-degree-of-freedom MMG motion model. Given that parameters are the core of the motion model and their computation is complex, the objective of SRUKF is to utilize collected data to determine the motion model parameters in real-time. The stability of the proposed algorithm is explored to demonstrate its applicability. Furthermore, it is evaluated using free-running model test data collected from the Maritime Research Institute Netherlands. The final prediction results are compared with the extended Kalman filter and UKF, which validated the superiority of the proposed scheme. Real-time parameter estimation is crucial for subsequent trajectory tracking control. The proposed control method can ensure a given tracking accuracy while satisfying practical applications. Online parameter identification research can also provide motion models for maritime simulators and be used to assess ship manoeuvring performance.

1. Introduction

1.1. Background and motivation

The manoeuvring motion model of surface vessels (SVs) serves as the foundation and primary concern for researching the path following (Li et al., 2024a; Sutulo et al., 2002) and trajectory tracking control (Perera et al., 2012; Zhang et al., 2022a). Manoeuvring models are the core component of manoeuvring simulators that are essential for training navigators and, nowadays, can even be used in desktop applications (Varela and Guedes Soares, 2015). Therefore, the identification modelling of SV manoeuvring motion has been in the limelight. The design process for SV is relatively complex and needs to consider various motion characteristics such as manoeuvrability and seakeeping performance. Furthermore, some design and dimension parameters are proprietary, making it difficult to obtain accurate SV motion models through model testing (Sutulo and Soares, 2006). Hydrodynamic derivatives (HDs) computation largely relies on empirical formulas, which have uncertainties associated (Sutulo and Guedes Soares, 2019). This

method may lead to considerable errors for SVs with significant differences in Froude numbers. However, SVs with similar hull geometries usually operate within a limited range of Froude numbers. Nonlinear identification theory (Merwe and Wan, 2001; Perera et al., 2015) can obtain motion models applicable to practical applications without relying on design and dimension parameters. Therefore, it is necessary to utilize the nonlinear identification modelling method to obtain the manoeuvring motion models of SV.

1.2. Related work

Based on the motion characteristics of SV, three ship manoeuvring models are generally available (Song et al., 2022; Sutulo and Guedes Soares, 2011), namely the Abkowitz, MMG, and Fossen models. The Abkowitz model had many HDs, and its physical meaning was controversial. The Fossen model initially used forces and moments as inputs for the SV motion model. However, forces and moments cannot be accurately monitored in the motion process of SVs. The MMG model has clear physical significance and comprehensively considers the effects of the

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rudder and propeller on SV motion. The model obtained is in line with practical maritime applications (Zhang et al., 2022b). Therefore, this study conducts corresponding identification modelling and trajectory tracking control research based on this model.

Extended Kalman Filter (EKF) has demonstrated promising results in many research fields (Davis and Fabien, 2020), including for parameter estimation (Xie et al., 2019) and navigational trajectory prediction (Perera et al., 2012). Compared to the EKF, the UKF has certain advantages in dealing with nonlinear systems and uncertainty. Zheng et al. (2022) obtained a simplified MMG model, where the nine hydrodynamic parameters of the model were identified. The final state estimation results demonstrated that UKF is more accurate than EKF. Compared with reference (Zheng et al., 2022), the motion model of the 6-degree-of-freedom (DOF) autonomous underwater vehicle established in reference (Sabet et al., 2018) was more complex. Sabet et al. (2018) took the parameter identification of the complex motion model as an example, similarly validating the superiority of UKF. In addition to the research results mentioned above, UKF has been widely applied in automobile, target tracking, and parameter identification modelling research (Liu et al., 2022; Muhammad and Ahsan, 2020). However, due to the SV motion model's complexity, few studies were conducted using UKF in the research on the identification modelling of SV motion. UKF also needs to be improved in terms of algorithm stability.

As research continues to advance and theoretical knowledge deepens, despite the positive research results obtained by UKF, it has been found to have certain limitations Zhao and Mili (2019); Ampountolas (2023) designed a commercial adaptive cruise control system with a constant time advance strategy and utilized the dual UKF to perform nonlinear state estimation and system parameter identification. Dang et al. (2022) proposed a weighted measurement model and combined it with the SRUKF to obtain dynamic state estimation of the power system. To address the issue of state estimation under unknown noise conditions, Zhang et al. (2022c) proposed an enhanced adaptive UKF and applied it to estimate the 3 DOF motion state of cars under time-varying noise conditions. In summary, these improved algorithms can enhance the applicability and accuracy of UKF (Wang et al., 2022). They have a beneficial effect on both system state estimation and parameter identification. Due to the coupling and strong nonlinearity of SV's motion, few improved algorithms are applied to the parameter identification of SV motion models compared with other research fields.

Trajectory tracking control requires establishing a plane motion model. Abrougui et al. (2021) integrated identification modelling with SV motion control. However, the parameter identification results lacked generalization verification, and the identified motion model was relatively simple. Different methods for parameter identification have been developed based on support vector machines (Wang et al., 2019; Zhu et al., 2020), allowing even real-time estimation (Xu et al., 2019; Pei et al., 2023). Once an accurate motion model is established or identified, adaptive control (Perera and Guedes Soares, 2013; Li et al., 2024; Xu et al., 2021), sliding mode control (SMC) (Perera and Guedes Soares, 2012; Li and Bu, 2021; Zhang et al., 2024), model predictive control (Han and Zhang, 2022), fuzzy control (Chen et al., 2024), neural network control (Deng et al., 2022), and prescribed performance control (Shen et al., 2024) can be further employed to verify the controllability of the model. PID and fuzzy control (Zhang et al., 2023) do not require a precise mathematical model; however, their anti-disturbance capabilities are relatively limited. Adaptive control and adaptive SMC have significant robustness against system parameter variations and external disturbances, making them well-suited for complex nonlinear systems (Zhang et al., 2025). Model predictive control can handle input-output constraints, making it well-suited for integrated tasks of complex path planning and trajectory tracking. Neural network control (Li et al., 2024b) addresses complex nonlinear problems in a data-driven manner, offering powerful modelling capabilities, particularly for scenarios with significant dynamic disturbances and high uncertainty. Prescribed performance control achieves high-precision trajectory tracking by

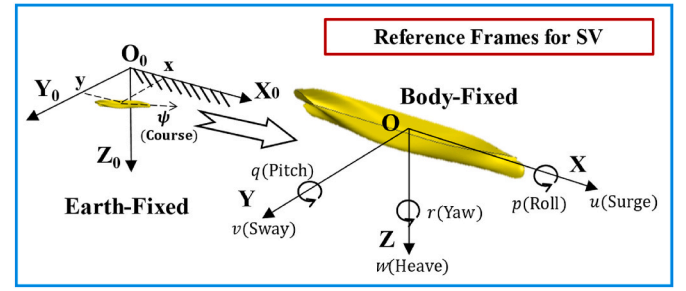


Fig. 1. 3-D coordinate systems of the SV.

predefining convergence performance metrics for trajectory errors, such as error bounds and convergence rates. The verification of controllability mainly considers whether the control objectives can be finished and applied to practical applications.

Given the above research work, SRUKF can make up for some limitations of UKF, and its stability in the parameter identification of SV motion model has not been proven. Compared to references (Xie et al., 2019; Zheng et al., 2022; Ampountolas, 2023; Wang et al., 2022; Zhang and Zhu, 2021), the MMG model is more complex and requires more parameters to be identified. This is also a challenge for the proposed scheme. In addition, few control schemes considering practical maritime applications are proposed. Whether it is state estimation or motion control, an accurate mathematical model is essential. There is a lack of research on the organic combination between online identification modelling and motion control.

1.3. Contributions and organization

Statement of Contributions: Establishing mathematical models is an effective way to study the performance of a system. To address the aforementioned challenges, this paper integrates the real-time nonlinear identification modelling of SV motion with trajectory tracking control. The main research work includes the following three aspects: (a) It first uses stochastic process theory to presents a stability analysis of SRUKF applied to SV motion identification modelling, and obtains a parameter identification model by Euler forward difference discretization. (b) It develops a SRUKF for the online nonlinear identification of discretized MMG model using free-running model test data. (c) It conducts trajectory tracking control of SV based on the identified model and adaptive SMC (ASMC). The proposed controller meets the requirements of practical applications.

Organization: Section 2 introduces the two coordinate systems, 3 DOF MMG motion model, discretized parameter identification model, and description and stability analysis of the SRUKF. Based on the model and algorithm established in Section 2, Section 3 presents the motion data, and parameter identification of the 3 DOF MMG model, and verifies the generalization of the identified parameters. Based on the identified motion model in Section 3, Section 4 utilizes ASMC to perform trajectory tracking control research. Section 5 provides the conclusions and prospects for future research.

2. System description and preliminaries

2.1. System description

SV motion can be described using both earth-fixed and body-fixed coordinate systems, which can represent different motion states of the SV. Fig. 1 shows the 3-D coordinate systems of the SV with earth-fixed and body-fixed frames. The earth-fixed coordinate system can describe the position and course of the SV, while the body-fixed coordinate system can describe the motion states.

Considering the mathematical model required for trajectory tracking

control of SV, a 3 DOF MMG motion model is established based on two coordinate systems. This model is widely used in SV motion control, i.e., it can accurately describe the motion while also meeting the requirements of motion control research. The motion model is expressed as Eq. (1). The stability and accuracy of this model have been verified in a previous study (Ogawa and Kasai, 1978).

$$\begin{cases} (m + m_x)\dot{u} - (m + m_y)vr = X_H + X_P + X_R + X_D \\ (m + m_y)\dot{v} + (m + m_x)ur = Y_H + Y_P + Y_R + Y_D \\ (I_{zz} + J_{zz})\dot{r} = N_H + N_P + N_R + N_D \end{cases} \quad (1)$$

where, m is the mass, m_x and m_y are added masses, u , v , and r are surge velocity, sway velocity, and yaw rate, respectively, I_{zz} and J_{zz} are the inertia moment and added inertia moment, the subscripts H , P , R , and D on X , Y , and N denote the longitudinal force, lateral force, and yaw moment experienced by the hull, propeller, rudder, and external environmental disturbance respectively.

Identification modelling is employed to eliminate the complex mechanistic calculations. The goal is to create a model that accurately reflects ship motion and aligns well with motion control. This paper adopts the method from reference (Son and Nomoto, 1981), which summarizes the effect of the open-water rudder and propeller on ship motion into two hydrodynamic coefficients, respectively. Without considering external environmental disturbances, the right-hand side of Eq. (1) can be represented as X , Y , and N , and expanding them obtains

$$\begin{cases} X = X_{uu}u^2 + X_{vv}v^2 + X_{rr}r^2 + X_{nn}n^2 + X_\delta\delta \\ Y = Y_vv + Y_r r + Y_{|v|v}|v|v + Y_{|r|r}|r|r + Y_{|v|r}|v|r + Y_\delta\delta \\ N = N_vv + N_r r + N_{|v|v}|v|v + N_{vv}v^2r + N_{rr}r^2 + N_\delta\delta \end{cases} \quad (2)$$

where, n denotes the propeller speed, δ represents the rudder angle. They are input variables of the SV motion model. The subscripted X , Y , and N represent HDs. For instance, $X_{uu} = \frac{\partial^2 X}{\partial u^2}$, represents the second-order derivative coefficient of the longitudinal force X with respect to the surge velocity u . It indicates the rate at which X changes with u^2 as u increases or decreases. $Y_\delta = \frac{\partial Y}{\partial \delta}$, represents the first-order derivative of the lateral force Y with respect to the rudder angle δ , describing the influence of rudder angle variations on the SV's lateral hydrodynamic forces.

The propeller thrust does not affect the lateral motion and yaw motion of the SV; it only affects the longitudinal motion. When steering, the rudder force generates components in three motion directions. Eq. (2) is plugged into Eq. (1). Utilizing the Euler forward difference discretization method, the parameter identification model is obtained as shown in

$$\left. \begin{aligned} \mathcal{Z}_{\mathcal{J}} &= \mathcal{F}_{\mathcal{J}}(u, v, r, \mathbf{U}) \\ \frac{\mathcal{Z}_{\mathcal{J}}(k+1) - \mathcal{Z}_{\mathcal{J}}(k)}{h} &= \boldsymbol{\theta}_{\mathcal{J}} \mathcal{Z}_{\mathcal{J}}(k) + \boldsymbol{\theta}_{\mathcal{J}\mathbf{U}} \mathbf{U}(k), \mathcal{J} = u, v, r \end{aligned} \right\} \quad (3)$$

where the variables are defined as shown in

$$\left. \begin{aligned} \mathcal{Z}(k+1) &= [u(k+1), v(k+1), r(k+1)]^T \\ \mathcal{Z}_u(k) &= [u(k), u^2(k), v(k)r(k), v^2(k), r^2(k)]^T \\ \mathcal{Z}_v(k) &= [v(k), r(k), |v(k)v(k)|, |r(k)r(k)|, |v(k)r(k)|, -u(k)r(k)]^T \\ \mathcal{Z}_r(k) &= [v(k), r(k), |v(k)v(k)|, v^2(k)r(k), v(k)r^2(k)]^T \\ \mathbf{U} &= [n^2, \delta]^T \\ \boldsymbol{\theta}_u &= [0, a_1, \dots, a_4]; \boldsymbol{\theta}_v = [b_1, b_2, \dots, b_6]; \boldsymbol{\theta}_r = [c_1, c_2, \dots, c_5] \\ \boldsymbol{\theta}_{u\mathbf{U}} &= [a_5, a_6]; \boldsymbol{\theta}_{v\mathbf{U}} = [0, b_7]; \boldsymbol{\theta}_{r\mathbf{U}} = [0, c_6] \end{aligned} \right\} \quad (4)$$

where, $\mathcal{F}(\cdot)$ represents the nonlinear function regarding the SV motion states, k denotes the sampling point, h is the sampling interval, the subscripted a , b , and c are the model parameters to be identified, which are obtained through the transformation of the HDs. For example, $b_2 = \frac{Y_r}{m+m_y}$. $\mathcal{Z}$ is the output vector of the identification system, $\mathbf{U}$ is the input vector of the identification system, $\mathcal{Z}$ with subscripts denotes the nonlinear state vector composed of the SV motion states, $\boldsymbol{\theta}$ with sub-

scripts denotes the vector of model parameters, all of which are constants.

Remark 1. Since the identified model parameters are unknown, the parameter identification model is transformed into Eq. (3). The identified hydrodynamic parameters are processed and integrated into $\boldsymbol{\theta}$. When using the fourth-order Runge-Kutta method to solve the SV motion model, Eq. (3) is more concise compared to the original model. Additionally, this model is also convenient for the trajectory tracking control of SV.

This study aims to determine the parameters of the 3 DOF MMG motion model. The model parameters are iteratively updated as system state variables to meet the requirements of the proposed scheme. Let $\boldsymbol{\theta} = [\boldsymbol{\theta}_u, \boldsymbol{\theta}_{u\mathbf{U}}, \boldsymbol{\theta}_v, \boldsymbol{\theta}_{v\mathbf{U}}, \boldsymbol{\theta}_r, \boldsymbol{\theta}_{r\mathbf{U}}]^T$. Finally, the established parameter identification model is

$$\begin{bmatrix} \mathcal{Z}_{\mathcal{J}}(k+1) \\ \boldsymbol{\theta}(k+1) \end{bmatrix} = \begin{bmatrix} \mathcal{Z}_{\mathcal{J}}(k) + h(\boldsymbol{\theta}_{\mathcal{J}}(k)\mathcal{Z}_{\mathcal{J}}(k) + \boldsymbol{\theta}_{\mathcal{J}\mathbf{U}}(k)\mathbf{U}(k)) \\ \boldsymbol{\theta}(k) \end{bmatrix} \quad (5)$$

The measurement equation of the identification system is

$$\mathcal{Y}_{k+1} = \mathbf{C}\boldsymbol{\xi}_{k+1} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \end{bmatrix} \boldsymbol{\xi}_{k+1} = [u_{k+1}, v_{k+1}, r_{k+1}]^T \quad (6)$$

The state vector of the nonlinear system is represented by $\boldsymbol{\xi}_k = [\mathcal{Z}_{\mathcal{J}}(k); \boldsymbol{\theta}(k)]$. The nonlinear state vector is represented by $\boldsymbol{\varsigma}_k = [\mathcal{Z}_{\mathcal{J}}(k); \boldsymbol{\theta}(k)]$. For convenience, the parameter identification model is transformed into a nonlinear function as $\boldsymbol{\xi}_{k+1} = F(\boldsymbol{\varsigma}_k, \mathbf{U}_k)$.

2.2. Square root unscented Kalman filter

SRUKF employs orthogonal triangle decomposition to handle the issue of computing the square root of the covariance matrix to enhance stability, making it applicable to various scenarios (Merwe and Wan, 2001). Its recursive process and fundamental framework are based on UKF. To introduce the proposed approach and demonstrate its stability, consider the discrete-time state and measurement models as

$$\begin{cases} \boldsymbol{\xi}_{k+1} = F(\boldsymbol{\varsigma}_k, \mathbf{U}_k) + \boldsymbol{\omega}_k \\ \mathcal{Y}_{k+1} = \mathbf{H}_{k+1}\boldsymbol{\xi}_{k+1} + \boldsymbol{\gamma}_{k+1} \end{cases} \quad (7)$$

where, $\boldsymbol{\xi}_k \in \mathcal{R}^{n_x}$, is the state vector, $\mathcal{Y}_k \in \mathcal{R}^{n_y}$, is the measurement vector, n_x and n_y represent the dimensions of the state and measurement vectors, $F(\cdot)$ is the nonlinear function of the motion state, $\mathbf{H}$ is the measurement matrix, $\boldsymbol{\omega}$ is the state noise, and $\boldsymbol{\gamma}$ is the measurement noise.

Remark 2. The state equation is nonlinear. There is a difference between the system state vector and the measurement vector, i.e., $\boldsymbol{\varsigma}_k \neq \boldsymbol{\xi}_k$. $\boldsymbol{\varsigma}_k$ represents the nonlinear terms related to the $\boldsymbol{\xi}_k$.

The main objective of SRUKF is to estimate the value of $\boldsymbol{\xi}_{k+1}$ at the next time step using the nonlinear state vector $\boldsymbol{\varsigma}_k$ from the previous time step. The estimated parameters will converge to a stable value as the recursion progresses. Select the initial system state $\boldsymbol{\xi}_0$ and square root of state covariance matrix $\mathbf{S}_0$. Based on the model established in Eq. (7), we select $2n_x + 1$ sigma points denoted as $\boldsymbol{\chi}$, as shown in

$$\left. \begin{aligned} \boldsymbol{\chi}_{k|k}^{(i)} &= \widehat{\boldsymbol{\xi}}_{k|k}, i = 0 \\ \boldsymbol{\chi}_{k|k}^{(i)} &= \widehat{\boldsymbol{\xi}}_{k|k} + \sqrt{(n_x + \lambda)}\mathbf{S}_k, i = 1, 2, \dots, n_x \\ \boldsymbol{\chi}_{k|k}^{(i)} &= \widehat{\boldsymbol{\xi}}_{k|k} - \sqrt{(n_x + \lambda)}\mathbf{S}_k, i = n_x + 1, n_x + 2, \dots, 2n_x \\ \lambda &= \alpha^2(n_x + \kappa) - n_x \end{aligned} \right\} \quad (8)$$

where, $\mathbf{S}$ is the square root of covariance matrix, $\boldsymbol{\chi}_{k|k}^{(i)}$ represents the i -th sigma point at the k -th step. λ is the scale factor, κ is the second scale factor, and α is a primary control factor.

Substituting χ into the state equation of Eq. (7), we get the predicted mean and square root of covariance matrix of the system state as shown in

$$\left. \begin{aligned} \hat{\xi}_{k+1|k} &= \sum_{i=0}^{2n_x} \mathbf{W}_m^{(i)} F(\zeta(\chi_{k|k}^{(i)}), \mathbf{U}_k) \\ \mathbf{S}_{k+1|k}^- &= \text{qr} \left\{ \left[\sqrt{\mathbf{W}_c^{(i)}} \left(\hat{\xi}_{k+1|k}^{1:2n_x} - \hat{\xi}_{k+1|k} \right), \sqrt{\mathbf{Q}_k} \right] \right\} \\ \mathbf{S}_{k+1|k} &= \text{cholupdate} \left\{ \mathbf{S}_{k+1|k}^-, \hat{\xi}_{k+1|k}^0 - \hat{\xi}_{k+1|k}, \mathbf{W}_c^{(0)} \right\} \\ \mathbf{W}_m^{(i)} &= \frac{\lambda}{n_x + \lambda}, i = 0 \\ \mathbf{W}_c^{(i)} &= \frac{\lambda}{n_x + \lambda} + (1 - \alpha^2 + \epsilon), i = 0 \\ \mathbf{W}_m^{(i)} &= \mathbf{W}_c^{(i)} = \frac{1}{2(n_x + \lambda)}, i = 1, 2, \dots, 2n_x \end{aligned} \right\} \quad (9)$$

where, $\text{qr}(\cdot)$ is orthogonal triangle decomposition, $\text{cholupdate}(\cdot)$ is Cholesky decomposition, $\mathbf{Q}_k$ is the state noise covariance matrix, $\mathbf{W}_m$ and $\mathbf{W}_c$ are the weight coefficients, ϵ is a constant containing the prior information of the probability distribution of ξ .

The χ is plugged into the measurement equation of Eq. (7). The predicted mean and square root of covariance matrix of the system measurement state are obtained as

$$\left. \begin{aligned} \hat{\mathcal{Y}}_{k+1|k} &= \sum_{i=0}^{2n_x} \mathbf{W}_m^{(i)} \mathbf{H}_k \chi_{k|k}^{(i)} \\ \mathbf{S}_Z^- &= \text{qr} \left\{ \left[\sqrt{\mathbf{W}_c^{(i)}} \left(\hat{\mathcal{Y}}_{k+1|k}^{1:2n_x} - \hat{\mathcal{Y}}_{k+1|k} \right), \sqrt{\mathbf{R}_k} \right] \right\} \\ \mathbf{S}_Z &= \text{cholupdate} \left\{ \mathbf{S}_Z^-, \hat{\mathcal{Y}}_{k+1|k}^0 - \hat{\mathcal{Y}}_{k+1|k}, \mathbf{W}_c^{(0)} \right\} \\ \mathbf{S}_{XZ} &= \sum_{i=0}^{2n_x} \mathbf{W}_c^{(i)} \left[\hat{\xi}_{k+1|k}^{1:2n_x} - \hat{\xi}_{k+1|k} \right] \left[\hat{\mathcal{Y}}_{k+1|k}^i - \hat{\mathcal{Y}}_{k+1|k} \right]^T \\ \mathbf{K}_k &= \frac{\mathbf{S}_{XZ}}{\mathbf{S}_Z \mathbf{S}_Z^T} \\ \hat{\xi}_{k+1|k+1} &= \hat{\xi}_{k+1|k} + \mathbf{K}_k (\mathcal{Y}_{k+1} - \hat{\mathcal{Y}}_{k+1|k}) \\ \mathbf{S}_{k+1|k+1} &= \text{cholupdate} \left\{ \mathbf{S}_{k+1|k}, \mathbf{K}_k \mathbf{S}_Z, -1 \right\} \end{aligned} \right\} \quad (10)$$

where, $\mathbf{R}_k$ is the measurement covariance matrix, $\mathbf{K}_k$ is the Kalman gain matrix.

SRUKF is based on Bayesian theory and unscented transformation and replaces the approximate nonlinear function itself with the approximate posterior mean and covariance of the system state. This method has high accuracy and is easy to conduct.

2.3. Stability analysis

Based on the theorem of exponential boundedness in the mean square sense of stochastic processes, this paper proves the stability of the proposed algorithm under the premise of estimating bounded state errors (Liu and Xie, 2010). Let $\tilde{\xi}_k = \xi_k - \hat{\xi}_k$, $\tilde{\xi}_{k|k-1} = \xi_k - \hat{\xi}_{k|k-1}$. Expand ξ_k around $\hat{\xi}_{k-1}$ using Taylor series. From Eq. (7), leads to

$$\xi_k = F(\zeta(\hat{\xi}_{k-1}), \mathbf{U}_{k-1}) + \sum_{j=1}^{\infty} \frac{D_{\xi_{k-1}}^j F(\zeta(\hat{\xi}_{k-1}), \mathbf{U}_{k-1}) \tilde{\xi}_{k-1}^j}{j!} + \omega_k \quad (11)$$

where, $D_{\xi_{k-1}}^j$ is the j -th derivative of ξ_k at point $\hat{\xi}_{k-1}$.

According to Eqs. (8), (9) and (11), an approximation can be obtained:

$$\tilde{\xi}_{k|k-1} \approx \mathbf{F}_k \zeta(\hat{\xi}_{k-1}) + \omega_k \quad (12)$$

To address the state prediction errors in Eq. (12) (Xiong et al., 2006), a diagonal matrix β_k is introduced, leading to

$$\tilde{\xi}_{k|k-1} = \beta_k \mathbf{F}_k \zeta(\hat{\xi}_{k-1}) + \omega_k = \mathbf{F}_k^\beta \zeta(\hat{\xi}_{k-1}) + \omega_k \quad (13)$$

The covariance matrix at step $k-1$ can be written as

$$\left. \begin{aligned} \mathbf{P}_{k|k-1} &= E \left\{ \tilde{\xi}_{k|k-1} \tilde{\xi}_{k|k-1}^T \right\} = \mathbf{F}_k^\beta \hat{\mathbf{P}}_{k-1} (\mathbf{F}_k^\beta)^T + \Delta \mathbf{P}_{k|k-1} + \mathbf{Q}_k \\ \hat{\mathbf{P}}_{k|k-1} &= \mathbf{F}_k^\beta \hat{\mathbf{P}}_{k-1} (\mathbf{F}_k^\beta)^T + \hat{\mathbf{Q}}_k \\ \hat{\mathbf{Q}}_k &= \Delta \mathbf{P}_{k|k-1} + \mathbf{Q}_k + \mathbf{P}^- + \Delta \mathbf{Q}_k \\ \Delta \mathbf{P}_{k|k-1} &= \mathbf{F}_k^\beta \hat{\mathbf{P}}_{k-1} (\mathbf{F}_k^\beta)^T - E \left(\mathbf{F}_k^\beta \tilde{\xi}_{k-1} \tilde{\xi}_{k-1}^T (\mathbf{F}_k^\beta)^T \right) \end{aligned} \right\} \quad (14)$$

where, $\mathbf{P}$ is covariance matrix, $\Delta \mathbf{Q}_k$ is an positive definite matrix, and $\mathbf{P}^- = \mathbf{F}_k^\beta \hat{\mathbf{P}}_{k-1} (\mathbf{F}_k^\beta)^T + \hat{\mathbf{Q}}_k - \mathbf{P}_{k|k-1}$.

Let $\tilde{\mathcal{Y}}_k = \mathcal{Y}_k - \hat{\mathcal{Y}}_k$. Accounting for Eqs. (7) and (10), leads to

$$\tilde{\mathcal{Y}}_{k|k-1} = \mathbf{H}_k \tilde{\xi}_{k|k-1} + \gamma_k \quad (15)$$

To prove the stability of the proposed algorithm, this paper introduces the definition and theorem of exponential boundedness in the mean square sense for stochastic processes, as shown in

$$E \{ \|\zeta_k\|^2 \} \leq \eta \|\zeta_k\|^{2i} + v, v > 0, 0 < i < 1, \forall k > 0 \quad (16)$$

$$\left. \begin{aligned} v_{\min} \|\varpi_k\|^2 &\leq V(\varpi_k) \leq v_{\max} \|\varpi_k\|^2, v_{\min}, v_{\max} > 0, \mu > 0, \forall k > 0 \\ E \{ V(\varpi_k) | \varpi_{k-1} \} - V(\varpi_{k-1}) &\leq \mu - \rho V(\varpi_{k-1}), 0 < \rho < 1 \\ E \{ \|\varpi_k\|^2 \} &\leq \frac{v_{\max}}{v_{\min}} E \{ \|\varpi_0\|^2 \} (1 - \rho)^k + \frac{\mu}{v_{\min}} \sum_{i=1}^{k-1} (1 - \rho)^i \end{aligned} \right\} \quad (17)$$

where, ζ_k and ϖ_k are both stochastic processes, $V(\varpi_k)$ is another stochastic process corresponding to ϖ_k . If Eqs. (16) and (17) are satisfied, then ζ_k and ϖ_k are said to be exponentially bounded in the mean square sense.

Based on Eqs. (7)–(10), (16) and (17), the estimation error of the proposed algorithm is said to be bounded in the mean square sense if the following theorem is satisfied:

$$\left. \begin{aligned} f_{\min}^2 \mathbf{I} &\leq \mathbf{F}_k (\mathbf{F}_k)^T \leq f_{\max}^2 \mathbf{I}, f_{\min}, f_{\max} \neq 0 \\ h_{\min}^2 \mathbf{I} &\leq \mathbf{H}_k \mathbf{H}_k^T \leq h_{\max}^2 \mathbf{I}, h_{\min}, h_{\max} \neq 0 \\ \beta_{\min}^2 \mathbf{I} &\leq \beta_k \beta_k^T \leq \beta_{\max}^2 \mathbf{I}, \beta_{\min}, \beta_{\max} \neq 0 \\ \hat{q}_{\min} \mathbf{I} &\leq \hat{\mathbf{Q}}_k \leq \hat{q}_{\max} \mathbf{I}, \mathbf{Q}_k \leq q_{\max} \mathbf{I}, \hat{q}_{\min}, \hat{q}_{\max}, q_{\max} > 0 \\ r_{\min} \mathbf{I} &\leq \mathbf{R}_k \leq r_{\max} \mathbf{I}, r_{\min}, r_{\max} > 0 \\ p_{\min} \mathbf{I} &\leq \mathbf{P}_k \leq p_{\max} \mathbf{I}, p_{\min}, p_{\max} > 0 \end{aligned} \right\} \quad (18)$$

where the subscripted f, h, β, q, r , and p are all real numbers, $\mathbf{I}$ denotes the unit matrix.

To prove that the proposed algorithm satisfies the theorem mentioned above, construct a Lyapunov function $V_k(\tilde{\xi}_k)$ as shown in

$$V_k(\tilde{\xi}_k) = \tilde{\xi}_k^T \mathbf{P}_k^{-1} \tilde{\xi}_k \quad (19)$$

Based on the conditions given in Eq. (18), Eq. (19) can be rewritten as $p_{\min}^{-1} \|\tilde{\xi}_k\|^2 \leq V_k(\tilde{\xi}_k) \leq p_{\max}^{-1} \|\tilde{\xi}_k\|^2$. According to Eq. (8) and the previous description, $\tilde{\xi}_k = \tilde{\xi}_{k|k-1} - \mathbf{K}_k \tilde{\mathcal{Y}}_k$. Eq. (19) can be rewritten as

$$V_k(\tilde{\xi}_k) = (\tilde{\xi}_{k|k-1} - \mathbf{K}_k \tilde{\mathcal{Y}}_k)^T \mathbf{P}_k^{-1} (\tilde{\xi}_{k|k-1} - \mathbf{K}_k \tilde{\mathcal{Y}}_k) \quad (20)$$

where, $\mathbf{K}_k$ are rearranged, $\mathbf{K}_k = \mathbf{P}_k \mathbf{H}_k^T \mathbf{R}_k^{-1}$, $\mathbf{P}_k^{-1} = \mathbf{P}_{k|k-1}^{-1} + \mathbf{H}_k^T \mathbf{R}_k^{-1} \mathbf{H}_k$.

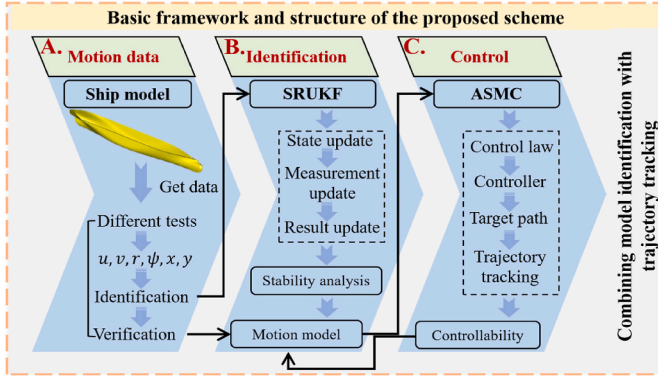


Fig. 2. Basic framework and structure of the proposed scheme.

To calculate the upper bound of $E\{V_k(\tilde{\xi}_k)|\tilde{\xi}_{k-1}\} - V_{k-1}(\tilde{\xi}_{k-1})$ in Eq. (17) (Xiong et al., 2006), leads to

$$\left. \begin{aligned} E\{V_k(\tilde{\xi}_k)|\tilde{\xi}_{k-1}\} - V_{k-1}(\tilde{\xi}_{k-1}) &\leq E\{(\mathcal{Z} - \mathcal{H} + \mathcal{V})|\tilde{\xi}_{k-1}\} - \mathcal{H} \\ \mathcal{Z} &= \omega_k^T \mathbf{g}^{-1} \omega_k \\ \mathcal{H} &= (\mathbf{H}_k \omega_k)^T [\mathbf{H}_k \mathbf{g} \mathbf{H}_k^T + \mathbf{R}_k]^{-1} (\mathbf{H}_k \omega_k) \\ \mathcal{V} &= \gamma_k^T \mathbf{R}_k^{-1} \mathbf{H}_k \mathbf{P}_k \mathbf{H}_k^T \mathbf{R}_k^{-1} \gamma_k \\ \mathcal{H} &= (\mathbf{H}_k \mathbf{F}_k^{\theta} \tilde{\xi}_{k-1})^T [\mathbf{H}_k \mathbf{g} \mathbf{H}_k^T + \mathbf{R}_k]^{-1} \mathbf{H}_k \mathbf{F}_k^{\theta} \tilde{\xi}_{k-1} \\ \mathbf{g} &= \mathbf{F}_k^{\theta} \mathbf{P}_{k-1} (\mathbf{F}_k^{\theta})^T + \hat{\mathbf{Q}}_k \end{aligned} \right\} \quad (21)$$

The μ in Eq. (17) has an upper bound given by

$$\mu_k \leq \text{tr}(\mathbf{g}^{-1} \mathbf{Q}_k) + \text{tr}[(\mathbf{R}_k^{-1} \mathbf{H}_k \mathbf{P}_k \mathbf{H}_k^T)] \leq \frac{q_{\max}}{q_{\min}} n_x + \frac{h_{\max}^2 p_{\max}}{r_{\min}} n_z \quad (22)$$

The q in Eq. (17) can be given by

$$\left. \begin{aligned} \mathcal{M} &= (\mathbf{H}_k \mathbf{F}_k^{\theta} \tilde{\xi}_{k-1})^T \mathbf{M} (\mathbf{H}_k \mathbf{F}_k^{\theta} \tilde{\xi}_{k-1}) \\ \mathbf{M} &= [(\mathbf{H}_k \mathbf{F}_k^{\theta} \mathbf{P}_{k-1} (\mathbf{H}_k \mathbf{F}_k^{\theta})^T + \mathbf{H}_k \hat{\mathbf{Q}}_k \mathbf{H}_k^T + \mathbf{R}_k)]^{-1} \\ q_k &= \mathcal{M} / (\tilde{\xi}_{k-1}^T \mathbf{P}_{k-1} \tilde{\xi}_{k-1}) \end{aligned} \right\} \quad (23)$$

According to Eqs. (17), (18) and (23), the q has a lower bound given by

$$p_{\min} (h_{\min} \beta_{\min} f_{\min})^2 [p_{\max} (h_{\max} \beta_{\max} f_{\max})^2 + \hat{q}_{\max} h_{\max}^2 + r_{\max}]^{-1} \leq q_k \quad (24)$$

Based on Eqs. (21)–(24), the definition and theorem related to Eqs. (16) and (17), we finally obtain the inequality as

$$E\{V(\tilde{\xi}_k)|\tilde{\xi}_{k-1}\} - V_{k-1}(\tilde{\xi}_{k-1}) \leq \mu_{\max} - q_{\min} V_{k-1}(\tilde{\xi}_{k-1}) \quad (25)$$

When Eq. (25) holds, then $E\{\|\tilde{\xi}_k\|^2\}$ has an upper bound as

$$E\{\|\tilde{\xi}_k\|^2\} \leq \frac{p_{\max}}{p_{\min}} E\{\|\tilde{\xi}_0\|^2\} (1 - q_{\min})^k + \frac{\mu_{\max}}{p_{\min}} \sum_{i=1}^{k-1} (1 - q_{\min})^i \quad (26)$$

3. Identification results

Section 2 introduces the identification model and verifies the stability of the proposed algorithm. Fig. 2 shows the basic framework and structure of the proposed scheme. This section develops the parameter identification of the SV motion model using free-running model test data.

3.1. Motion data of SV

The free-running model test is an efficient and safe method for assessing SV's performance. It can also provide data support for ship

Table 1
Main particulars of the SV.

Parameters	Values
Length between perpendiculars (L_{pp}/m)	6.0702
Breadth (B/m)	0.8498
Draft (d/m)	0.2850
Block coefficient ($C_B/-$)	0.6510
Displacement (∇/m^3)	0.9565

motion modelling. This paper collected SV motion data under various manoeuvring conditions. The diversity of these data sets can be used for identification modelling and generalization verification. The collected test data, primarily sourced from the Maritime Research Institute Netherlands (MARIN) during the [SIMMAN 2020](#) workshop (SIMMAN, 2020). MARIN is a leading institution with substantial influence in the fields of maritime technology and ship behavior analysis. Data pre-processing is required, including tasks such as data segmentation, to ensure that the dataset meets the requirements for system identification modelling. The data is then divided into identification and test samples. The propeller operated at a constant speed of approximately 642 RPM, with an initial SV speed of around 2 m/s. The tests were conducted in deep water under similar environmental conditions for each trial. The datasets encompass critical SV's motion scenarios, including turning and zigzag tests. These data include detailed motion states (position, velocity, and heading angle), propeller speed, and rudder angle. The main particulars of the research object are shown in Table 1.

Data quality is crucial for the nonlinear identification of ship manoeuvring motion. As the data used is obtained through real measurements, it inevitably contains some instrument measurement errors. However, the proposed scheme is an optimal recursive estimation algorithm particularly well-suited for handling dynamic system data with noise. By combining the system's mathematical model with actual observational data, it estimates the states while effectively suppressing the influence of noise. These different test data are used to validate the effectiveness of the proposed algorithm and the generalization of the identified parameters. The test data were recorded at 12.5 Hz (0.08 s) (SIMMAN, 2020). The selection of training sample data is critically important. Excessive training data can lead to overfitting, while insufficient motion data may fail to capture the nonlinear characteristics of SV motion. To accurately estimate model parameters and avoid overfitting, this study utilizes the partial 20°/20° zigzag test as the identification sample (Zhang et al., 2015). The 20°/20° zigzag test is more suitable for studying dynamic maneuvering characteristics, particularly under normal operating conditions, as it allows for the analysis of the SV's response speed, course stability, and dynamic maneuvering performance. This test involves multiple alternating rudder angle changes, effectively containing the nonlinear motion characteristics of the SV.

The test samples include the −10°/−10° zigzag test and the −35° turning test data. The −10°/−10° zigzag test features rudder angle excitations opposite to and smaller than those of the identification sample, representing a small-amplitude maneuvering test. This test is particularly suited for evaluating the SV's sensitivity during small-angle maneuvers and its dynamic response performance under minor disturbance conditions. The −35° turning test involves rudder angle excitations that are larger than those in the identification sample and remain constant. This test is primarily used to evaluate the SV's steering and extreme maneuvering performance, emphasizing the hydrodynamic and hull interaction effects under large rudder angles. Therefore, these two test

Table 2
Motion data of SV.

Data	Test types	Number of data
Identification data	20°/20° Zigzag test	700
Test samples	−10°/−10° Zigzag test	1094
	−35° Turning test	1355

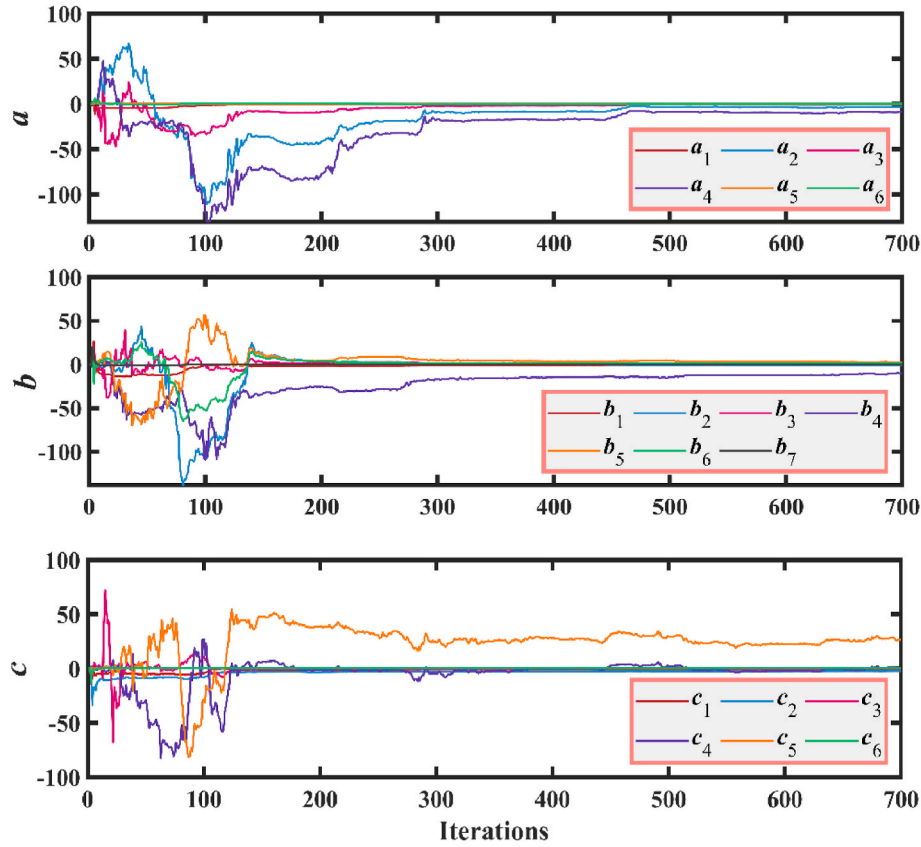


Fig. 3. Convergence results of motion model parameters.

data are selected as generalization validation samples in this study. The types, numbers, identification sample, and test samples of the trial data are shown in Table 2.

Remark 3. The number of data corresponds to the number of sampling points. For instance, there are 700 identification data with a sampling time of 56s.

3.2. Parameter identification of motion model

After selecting the identification model and data of SV, some parameters of the proposed scheme, such as the initial covariance matrix, are set. In the algorithm settings, $\omega = \text{diag}(10^{-3}, 5 \times 10^{-3}, 5 \times 10^{-3}, \mathbf{0}_{1 \times 19})$, $\gamma = \text{diag}(10^{-2}, 10^{-2}, 10^{-2})$, $\xi_0 = [u(0), v(0), r(0), \mathbf{0}_{1 \times 19}]^T$, $\mathbf{Q} = \omega^2$, $\mathbf{R} = \gamma^2$, $\mathbf{S}_0 = \mathbf{I}_{19 \times 19}$. The number of identification data is 700. The model parameters to be estimated are continuously updated during the iterative process, eventually converging to stable values. Additionally, based on the relationship between the coordinate systems illustrated in Fig. 1, the ψ , motion trajectory x and y of the SV can be expressed using

$$\left. \begin{aligned} \dot{\psi} &= r \\ \dot{x} &= u \cos(\psi) - v \sin(\psi) \\ \dot{y} &= u \sin(\psi) + v \cos(\psi) \end{aligned} \right\} \quad (27)$$

Since the parameters of the motion model are all unknown, this study aims to utilize the identification data to estimate the model parameters. The remaining test data is employed for generalization prediction. Furthermore, the Mean Square Error (MSE) is utilized as a criterion for evaluating the accuracy of the predicted results. It can also demonstrate the effectiveness of the proposed approach. Fig. 3 illustrates the convergence results of the identified parameters.

From the convergence results in Fig. 3, it can be observed that, except for the parameter c_5 which fluctuates during the iteration process. However, the fluctuation amplitude of c_5 is small, and it fluctuates

slightly around a stable value. The other parameters all converge to a stable value. In Section 2.2, we demonstrate the stability of the proposed algorithm. Additionally, Fig. 3 further illustrates the feasibility of applying the proposed algorithm to the nonlinear identification of the motion model. The SV motion is significantly nonlinear and complex. Therefore, the generalization results are crucial for evaluating the parameters' accuracy.

Data selection is crucial for nonlinear system identification modeling. Zigzag test data contain more nonlinear motion characteristics of the SV and are widely used as identification data (Sutulo and Guedes Soares, 2023; Xu et al., 2020). Based on the identified parameters and Eqs. (1)–(4), the prediction results of the $-10^\circ/-10^\circ$ zigzag test data are obtained using the fourth-order Runge-Kutta integration method. To validate the superiority, the obtained results are compared with multi-innovation least-squares, as presented in reference (Song et al., 2022). However, the prediction results obtained by this algorithm diverged. The proposed scheme has been verified on models with few parameters, such as the response model. The results obtained are compared with those obtained by EKF and UKF. The parameters of the comparison algorithms are set to be consistent with the proposed approach. The comparison results have shown the accuracy and efficiency of the proposed scheme. Fig. 4 shows the prediction results of $-10^\circ/-10^\circ$ zigzag test data. Table 3 presents the prediction accuracy assessed by MSE for $-10^\circ/-10^\circ$ zigzag test data.

The nonlinear characteristics of SV reflected by the $-10^\circ/-10^\circ$ zigzag test data and the identification data are different. As shown in Fig. 4, based on the identified parameters, the prediction results are consistent with the free-running model test data. Especially in the first 50 s, the degree of coincidence between the prediction data and the free-running model test data is very high. Fig. 4 demonstrates that the identified parameters are suitable for a zigzag test with a small rudder angle. To provide an overall comparison of the prediction accuracy, the MSEs of the predicted results for the three motion states are summarized. The

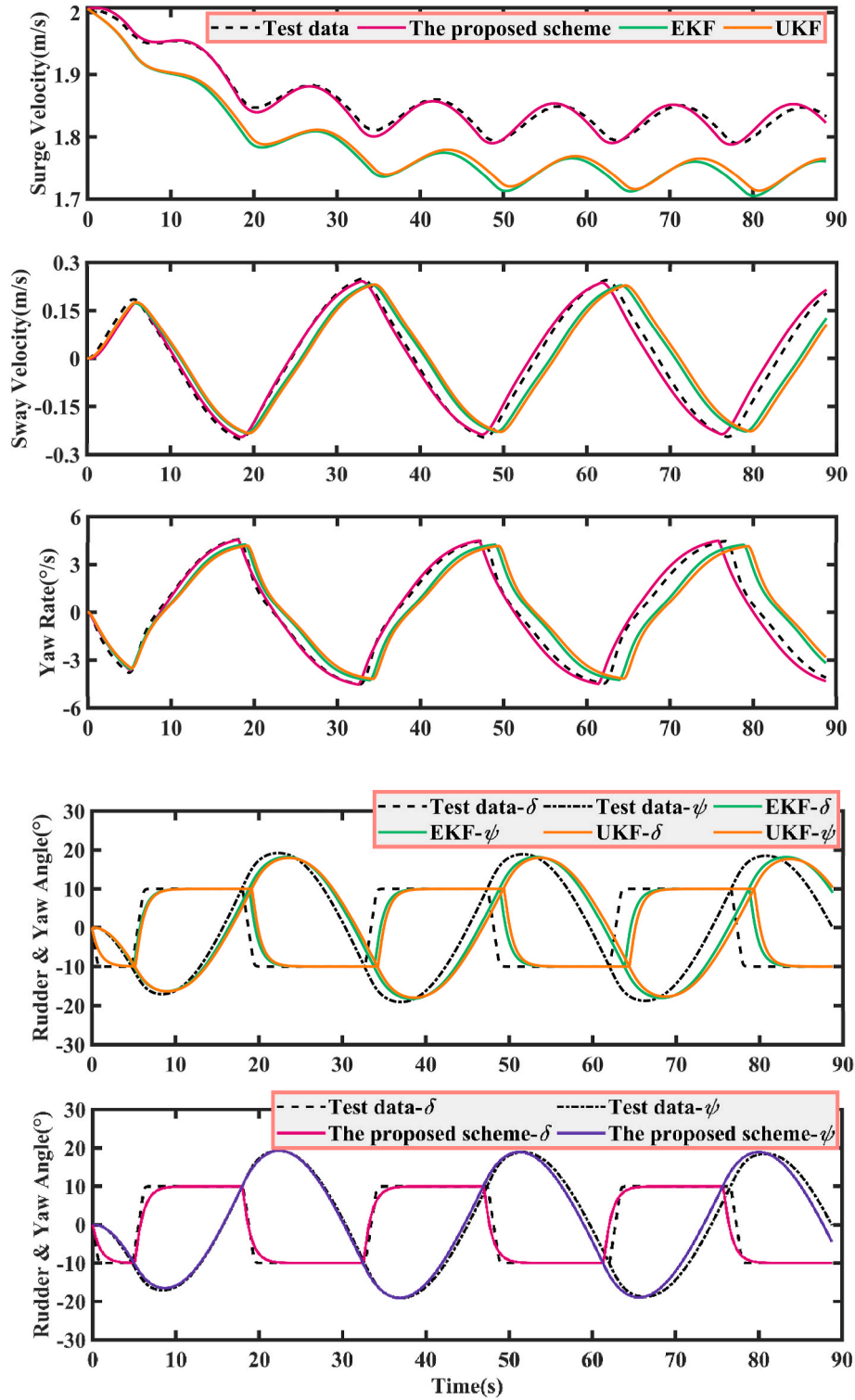


Fig. 4. Prediction results of $-10^\circ/-10^\circ$ zigzag test data.

Table 3

Prediction accuracy assessed by MSE for $-10^\circ/-10^\circ$ zigzag test data.

	u	v	r
The proposed scheme	4.38×10^{-5}	4.58×10^{-4}	0.15
EKF	0.0057	0.0030	1.08
UKF	0.0051	0.0047	1.68

proposed scheme achieves the highest accuracy, with the smallest MSE observed for surge velocity at 4.38×10^{-5} , while the yaw rate has the largest MSE at 0.15. In contrast, the UKF shows less accuracy, with the smallest MSE for sway velocity at 0.0047 and the largest MSE for yaw rate at 1.68. Similarly, the EKF demonstrates slightly improved accuracy compared to UKF but remains inferior to the proposed scheme, with the smallest MSE for sway velocity at 0.0030 and the largest MSE for yaw rate at 1.08. Overall, the proposed scheme outperforms both EKF and UKF in terms of prediction accuracy across the three motion states.

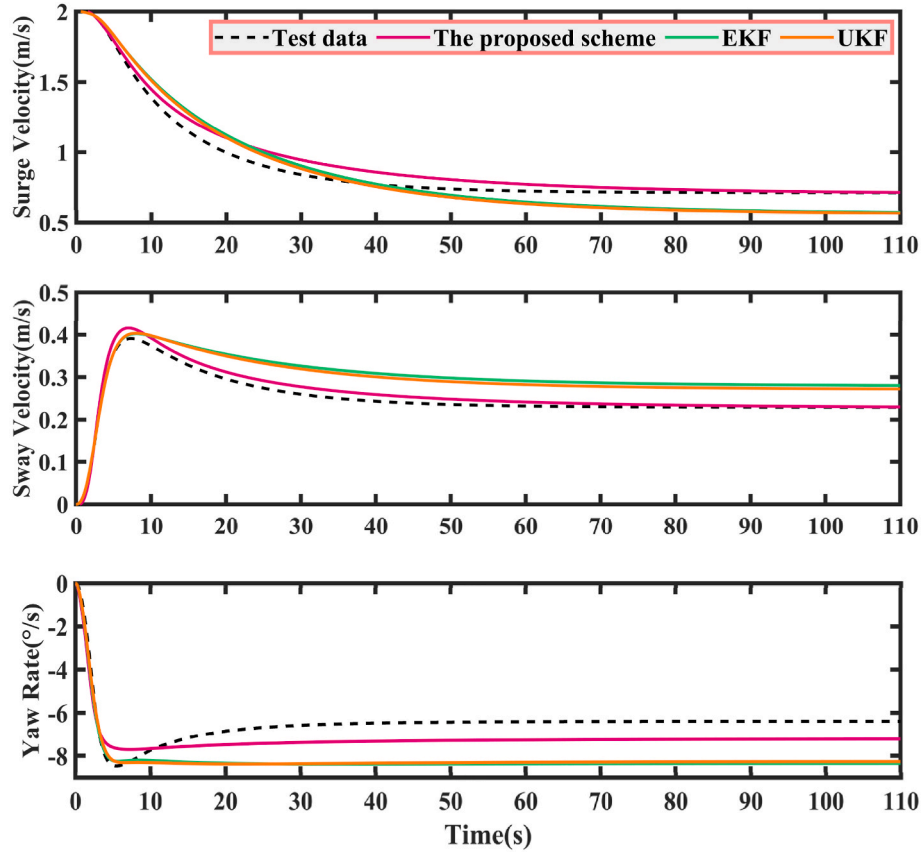
Fig. 5. Prediction results of -35° turning test data.

Table 4

Prediction accuracy assessed by MSE for -35° turning test data.

	u	v	r
The proposed scheme	0.0038	1.57×10^4	0.60
EKF	0.0103	0.0030	3.13
UKF	0.0106	0.0023	2.95

As a key evaluation criterion in the zigzag test, the first and second overshoot angles of the test data are 17.05° and 19.22° , respectively. For the proposed scheme, EKF, and UKF, the first and second overshoot angles are 16.51° and 19.27° , 16.30° and 18.18° , and 16.35° and 17.93° , respectively. Notably, the overshoot angles obtained by the proposed scheme are very close to the test data. The difference between the turning and zigzag tests is significant. It only steers once, focusing more on the SV's turning performance. To further validate the generalization of the identified parameters, this study shows the prediction results for the -35° turning test data, as illustrated in Fig. 5. Table 4 presents the prediction accuracy assessed by MSE for -35° turning test data.

Since the turning test does not clearly reflect the relationship between the yaw and rudder angles, it is not listed in the results. The overall prediction results of the proposed scheme, as shown in Fig. 5, indicate that while minor differences exist, the overall trend closely matches the test data. This further shows that the identified parameters suit a manoeuvre test with a large rudder angle. For the proposed scheme, the prediction result of yaw rate has the largest MSE at 0.60, while the sway velocity achieves the smallest MSE at 1.57×10^{-4} . In contrast, EKF and UKF show larger errors, with yaw rate MSEs of 3.13 and 2.95, respectively. The sway velocity achieves the smallest MSE at 0.0030 and 0.0023. These results highlight the superior accuracy of the proposed scheme, particularly in handling large rudder angle maneuvers. Based on Eq. (27) and the predicted motion states, the predicted

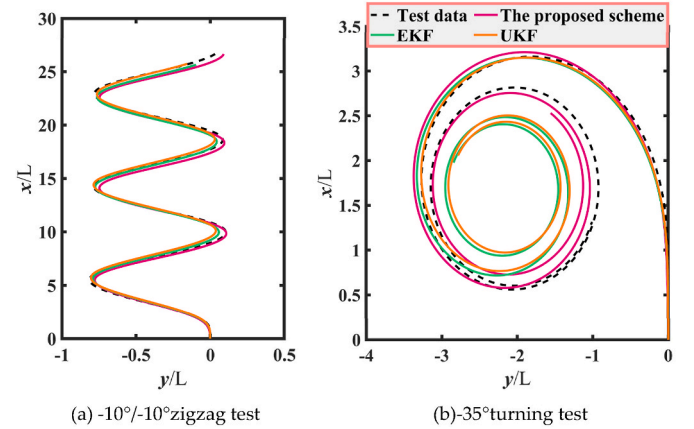


Fig. 6. Prediction results of trajectory.

trajectories for the two tests are shown in Fig. 6. From Fig. 6, it can be observed that the prediction accuracy for the $-10^\circ/-10^\circ$ zigzag test is higher than that for the -35° turning test. The accuracy of trajectory prediction can be reflected in the predicted motion state. Taking Fig. 6 (a) as an example, the surge velocity prediction results of the two comparison schemes are smaller than the test data. As a result, the predicted trajectories of the two comparison schemes are slower than the test data. In Fig. 6 (b), the turning radius predicted by the proposed scheme is closer to the test data. However, due to the slightly over-estimated surge velocity prediction, the motion trajectory is slightly faster than the test data.

This section uses partial $20^\circ/20^\circ$ zigzag test data to conduct the parameter identification of the 3 DOF manoeuvring motion model.

$-10^\circ/10^\circ$ zigzag and the -35° turning test data, as generalization validation samples, are utilized to demonstrate the effectiveness of the proposed approach and the generalization of the identified parameters. Additionally, the SRUKF, as a traditional identification modelling algorithm, consumes less time in the process of SV motion identification modelling. In the case of 700 identification samples, it only takes 0.12s to complete the iteration process, which also reflects the high efficiency of the proposed scheme in terms of time consumption.

4. Trajectory tracking control of SV

The 3 DOF maneuvering motion model of SV was identified in Section 3, which was widely used in SV course keeping (Yan et al., 2020) and trajectory tracking control (Li et al., 2024). When an SV motion model is unknown, corresponding research, such as SV motion control and simulation, will be impossible. In other words, practical applications, especially for SV, need to be based on a lot of theoretical research work. Establishing a solid theoretical foundation is essential to ensure the safety and efficacy of practical applications. Based on the verified SV motion model, this paper uses the ASMC to perform trajectory tracking control.

4.1. Controller design

Let the desired trajectory as (x_d, y_d) , position tracking errors as $x_e = x - x_d$, $y_e = y - y_d$. Then, $\dot{x}_e = \dot{x} - \dot{x}_d$, $\dot{y}_e = \dot{y} - \dot{y}_d$. Consider the identified model, such as Eq. (3), which has two system inputs, namely the n and δ . The rudder force exerted on the SV has a significant effect on longitudinal, lateral and yaw motion. However, the SV used in this paper is an underactuated vessel, with lateral force generated only when steering. Let $\tau_r = c_6\delta$. This paper first designs a controller for lateral motion. Then, based on this controller, designs a controller for longitudinal motion. In addition, although the identification data contains noise interference, time-varying disturbances are added to the motion model in this study to validate the robustness of the proposed controller. It is worth mentioning that the model in Eq. (3) is processed to parameters, such as $b_7 = \frac{Y_\delta}{(m+m_y)}$. Hence, three different types of time-varying disturbances, denoted as $d_u = \frac{x_p}{(m+m_x)} = 0.008\sin(0.2t)$, $d_v = \frac{y_p}{(m+m_y)} = 0.008\cos(0.5t)$, and $d_r = \frac{N_p}{(I_{zz}+J_{zz})} = 0.005\cos(0.5t)$ are introduced (Deng et al., 2022; Lei and Zhang, 2024).

Remark 4. These identified parameters have varying degrees of influence on the performance of the trajectory tracking controller. For instance, the identified parameter a_5 captures the second-order nonlinear effects of propeller speed on the SV's longitudinal thrust X , making it a critical parameter for studying ship motion modelling and control. Similarly, the identified parameter c_6 describes the sensitivity of yaw moment to rudder angle variations, serving as an essential parameter for evaluating the SV's manoeuvrability and control performance.

Let α_u and α_v as the virtual control variables for the u and v , respectively. The purpose is to use these two virtual control variables to indirectly control the motion trajectory of the SV. The α_u and α_v are shown in

$$\left. \begin{aligned} \alpha_u &= (\dot{x}_d - \rho x_e/w)\cos(\psi) + (\dot{y}_d - \rho y_e/w)\sin(\psi) \\ \alpha_v &= -(\dot{x}_d - \rho x_e/w)\sin(\psi) + (\dot{y}_d - \rho y_e/w)\cos(\psi) \\ w &= \sqrt{x_e^2 + y_e^2} + C \end{aligned} \right\} \quad (28)$$

where, $\rho > 0$, $C > 0$, are constants.

Combining Eqs. (27) and (28), leads to

$$\left. \begin{aligned} \dot{x}_e &= -\rho x_e/w + (u - \alpha_u)\cos(\psi) + (v - \alpha_v)\sin(\psi) \\ \dot{y}_e &= -\rho y_e/w - (u - \alpha_u)\sin(\psi) + (v - \alpha_v)\cos(\psi) \end{aligned} \right\} \quad (29)$$

As long as $(u - \alpha_u) \rightarrow 0$ and $(v - \alpha_v) \rightarrow 0$ in Eq. (29), the Lyapunov

function is constructed as $V = \frac{1}{2}x_e^2 + \frac{1}{2}y_e^2$. Then, $\dot{V} = -\rho(x_e^2 + y_e^2)/w \leq 0$. It can be inferred that the position trajectory errors x_e and y_e are asymptotically stable.

Define the sway velocity error as $v_e = v - \alpha_v$, $\Omega_v = b_7/c_6$. Then, the force exerted by the rudder in the lateral motion is $\Omega_v\tau_r$. Design the first-order sliding mode surface as

$$s_v = \sigma_{v1}v_e + \sigma_{v2} \int_0^t v_e(\tau) d\tau \quad (30)$$

and derive it to obtain

$$\dot{s}_v = \sigma_{v1}(\dot{v} - \dot{\alpha}_v) + \sigma_{v2}v_e \quad (31)$$

where, σ_{v1} and σ_{v2} are constants.

According to the identified model and three time-varying disturbances, $\dot{v} = \mathcal{F}_v(u, v, r) + \Omega_v\tau_r + d_v$. Let $\dot{s}_v = 0$. Then, the equivalent control law of lateral motion can be obtained as

$$\tau_{req} = \frac{1}{\Omega_v} \left[\dot{\alpha}_v - \mathcal{F}_v(u, v, r) - d_v - \frac{\sigma_{v2}}{\sigma_{v1}}v_e \right] \quad (32)$$

The switching control law is selected as

$$\tau_{rs} = -K_v \tanh(s_v / \varepsilon_v) \quad (33)$$

The final lateral motion control law is

$$\tau_r = \tau_{req} + \tau_{rs} \quad (34)$$

where, K_v and ε_v are constants.

According to the above introduction, it can be seen that $\delta = \tau_r/c_6$. Considering the manoeuvrability of SV, the servo and propeller speed needs to meet certain conditions, as shown in Eq. (35) (SIMMAN, 2020). Therefore, it is necessary to constrain τ_r . With the constraint of τ_r , it is more consistent with practical navigation, but the control efficiency may decrease.

$$\left. \begin{aligned} |\delta| &\leq 35^\circ \\ |\dot{\delta}| &= 14.28^\circ/s \\ 0 &\leq n \leq 642 \text{ rpm} \end{aligned} \right\} \quad (35)$$

Longitudinal motion is affected by propeller thrust and rudder force. But lateral motion is only affected by rudder force. Therefore, the first design is for the lateral motion control law. Then, the control law for longitudinal motion is designed. Let surge velocity error as $u_e = u - \alpha_u$, $\tau_u = a_5n^2$. Design the first-order sliding mode surface as

$$s_u = \sigma_{u1}u_e + \sigma_{u2} \int_0^t u_e(\tau) d\tau \quad (36)$$

and deriving Eq. (36), one obtains

$$\dot{s}_u = \sigma_{u1}(\dot{u} - \dot{\alpha}_u) + \sigma_{u2}u_e \quad (37)$$

where, σ_{u1} and σ_{u2} are constants.

Let $\Omega_u = a_6/c_6$, $\dot{s}_u = 0$. Then, $\dot{u} = \mathcal{F}_u(u, v, r) + \Omega_u\tau_r + \tau_u + d_u$. The equivalent control law of longitudinal motion is obtained as

$$\tau_{ueq} = \dot{\alpha}_u - \mathcal{F}_u(u, v, r) - \Omega_u\tau_r - d_u - \frac{\sigma_{u2}}{\sigma_{u1}}u_e \quad (38)$$

The switching control law is selected as

$$\tau_{us} = -K_u \tanh(s_u / \varepsilon_u) \quad (39)$$

The final control law of longitudinal motion is

$$\tau_u = \tau_{ueq} + \tau_{us} \quad (40)$$

where, K_u and ε_u are constants.

In the motion data of SV, the maximum propeller speed is shown in Eq. (35). Therefore, this paper does not consider reversing, that is, the propeller speed is always greater than 0, and sets $n \leq 642$ rpm. τ_u is also

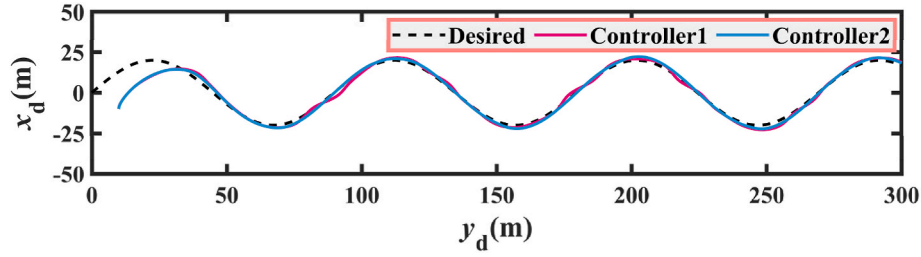


Fig. 7. Results of trajectory tracking control.

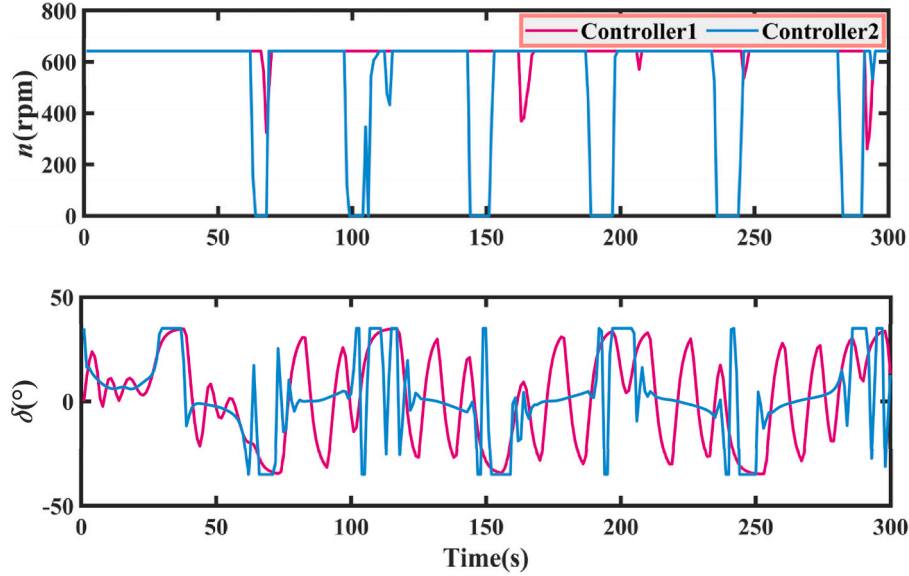


Fig. 8. Inputs of two controllers.

constrained accordingly.

Remark 5. This cannot be conducted accurately if the controller's inputs are forces and moments in practical applications. Therefore, they need to be converted into the rudder angle and propeller speed. Both inputs can be accurately set via the rudder and thruster devices.

4.2. Numerical experiment

Based on the above controller settings, this paper conducts trajectory tracking control of SV. This controller takes into account the manoeuvring characteristics of the SV and the servo system, which can meet the practical application. In addition, the free-running model test data contains certain noise interference. This means that the identified scheme considers interference when estimating the model parameters.

According to the manoeuvrability of SV, let $x_d = 20 \sin(0.07t)$, $y_d = t$, $x_0 = -10$, and $y_0 = 10$. The rest of the motion states are set to 0. Since the identification model already contains noise interference, this

paper does not add it to the trajectory tracking control. Furthermore, this paper aims to verify the controllability of the identified model. The designed controller needs to determine some design parameters. This paper sets $\rho = 1.5$, $C = 10$, $\sigma_{v1} = 2.5$, $\sigma_{v2} = 0.9$, $\sigma_{u1} = 1.4$, $\sigma_{u2} = 6$, $K_v = 0.1$, $K_u = 0.01$, $\varepsilon_v = 0.1$, and $\varepsilon_u = 0.1$.

Define the proposed controller as Controller1. For comparison, a controller that only constrains the amplitude of the rudder angle and propeller speed without constraining their rate is defined as Controller2. The parameter settings of the two controllers are the same. The results of trajectory tracking control are shown in Fig. 7. Fig. 8 compares the inputs of two controllers.

It can be seen from Fig. 7 that the control effect of Controller2 is better than that of Controller1. However, two inputs of Controller1 are suitable for practical applications. As can be seen from Fig. 8, two inputs of Controller2 change suddenly with a large amplitude. Controller1 adds corresponding constraints. The amplitude and rate of the two inputs are in line with the manoeuvring characteristics of SV. Therefore, it is necessary to consider not only the accuracy of the controller but also

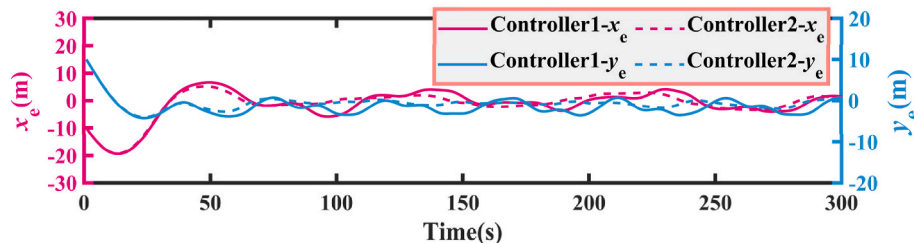


Fig. 9. Trajectory tracking errors of two controllers.

whether the input is satisfactory for practical applications. Fig. 9 shows the trajectory tracking errors of the two controllers. Although the proposed controller has specific errors during SV steering, it is more consistent with practical applications than Controller2.

Based on the identified manoeuvring motion model of SV and the proposed control scheme, this section conducts the trajectory tracking control of SV. A controller that only constrains the amplitude is used as a comparative scheme. Its limitations in practical maritime applications are presented. In contrast, the proposed control scheme fully considers the characteristics of the rudder and thruster and can meet practical navigation applications.

Remark 6. It is worth noting that the purpose of this research is first to use the nonlinear identification method to perform parameter identification of the unknown motion model. Then, the research of trajectory tracking control is conducted using the identified model. The application of trajectory tracking control also demonstrates the reliability and effectiveness of the identified model from another perspective.

5. Conclusions and outlook

Based on test data, system identification, and motion control theories, this study integrates online nonlinear identification modelling and trajectory tracking control. Based on the research results and discussion, the research work is mainly reflected in the following aspects:

- This study analyzes the stability of the proposed approach in the nonlinear identification modelling of the SV motion. The convergence of all 19 identified parameters under certain conditions further confirms its effectiveness. This demonstrates the feasibility of the proposed scheme in the application of SV nonlinear identification modelling and has good applicability.
- Based on these identified parameters and 4th-order Runge-Kutta integration, the coincidence degree between the predicted motion states and trajectory and the free-running model test data is high. Under the premise of less time consumption, the MSE of the motion prediction results is less than 0.60. Compared to EKF and UKF, the proposed scheme demonstrates higher prediction accuracy in validating both small and large rudder angle test data. This verifies the generalizability and superior of the proposed scheme and that the results obtained can be applied to research related to SV motion simulation and control.
- Based on the identified model, the ASMC is utilized to perform the trajectory tracking control. This study develops a controller that ensures accuracy and is suitable for practical applications.

This controller's successful design further demonstrates the identified model's reliability.

However, although the proposed nonlinear identification and motion control schemes have some good results, both schemes still need further to be improved. For instance, the proposed control scheme need to further improve trajectory tracking accuracy. In future research, the proposed scheme will be further applied to the unmanned SV in the laboratory, conducting an organic combination of identification modelling and motion control in practical applications.

CRediT authorship contribution statement

Yao Meng: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Xianku Zhang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition. **Xiufeng Zhang:** Writing – review & editing, Supervision, Methodology. **Yating Duan:** Writing – review & editing, Supervision. **C. Guedes Soares:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. The identified parameters for the mathematical model of SV

The identified parameters of the SV motion mathematical model are listed in Table 5. These parameters are combinations of HDs.

Table 5
Identification results for the SV in Section 3

The Identified Parameters of Motion Model		
$a_1 = -0.023$	$b_1 = -0.8553$	$c_1 = -0.7215$
$a_2 = -3.6141$	$b_2 = -0.4944$	$c_2 = -2.4756$
$a_3 = -0.8381$	$b_3 = 0.4451$	$c_3 = 0.1106$
$a_4 = -9.0068$	$b_4 = -9.9044$	$c_4 = 0.1936$
$a_5 = 0.0008$	$b_5 = 2.0548$	$c_5 = 26.5057$
$a_6 = -0.0012$	$b_6 = 1.0348$	$c_6 = 0.4045$
–	$b_7 = 0.1306$	–

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